Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/16/2005
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Project Title:	High Pressure Studies of Carbon Monoxide Oxidation for Application to Mars Exploration		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60607-7021	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	11/18/2003	**End Date:**	11/17/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/17/2007 per PI (4/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GA20G		
Performance Goal No.:			
Performance Goal Text:			
	The investigation of the high pressure combustion chemistry of carbon monoxide is proposed for application to the rocket engine propulsion of a Mars surface vehicle. The rocket engine will be fueled with carbon monoxide and oxygen produced in-situ from the electrolysis of Martian atmospheric carbon dioxide. In-situ production has the advantage of eliminating the need to transport fuel and oxidizer from Earth but has, in the case of a carbon monoxide/oxygen rocket engine, the disadvantage of generating a fuel with relatively low performance in the rocket engine. Consequently, the maximum performance possible from a carbon monoxide engine is necessary and optimal engine design is required to achieve it. Part of the engine design involves an evaluation of the effect of finite rate oxidation chemistry of carbon monoxide on performance. The evaluation of the chemistry necessitates a detailed chemical kinetic model that for the best engine design needs to capture very accurately all the chemical steps that influence the combustion of carbon		

Task Description:	monoxide. There are detailed chemical kinetic models available that do include the most current understanding of carbon monoxide oxidation but they have never been validated by experiments conducted at the operating pressures of the proposed Martian vehicle rocket engines: 200-300 psi (13.7-20.4 atm) for pressure fed engines, 2000-3000 psi (136-204 atm) for turbopump fed engines. These high pressures are rarely achievable in laboratory experiments that investigate chemical kinetics but are routinely accessed in the very high pressure single pulse shock tube operating at the University of Illinois at Chicago. This shock tube can create pressure ranges from 5-1000 atm, temperatures currently from 950-1450 K but extendable to 1900K and reaction times from 0.5-2.5 msec. These are the ranges of conditions that will used to experimentally examine the bone dry and hydrogen containing oxidation of carbon monoxide. Gas chromatography/mass spectrometry/thermal conductivity measurements of the carbon monoxide, oxygen, carbon dioxide, water species concentrations as a function of pressure, temperature, hydrogen content or water content, and stoichiometry can be compared with the predictions from existing models. Prior work examining the comparison between predictions of existing models for hydrocarbon oxidation and high pressure results obtained from the shock tube have shown that hydrocarbon models developed and validated against lower pressure experiments require modification in order to predict high pressure results. Therefore the hypothesis underlying the proposed experimental examination of the oxidation of carbon monoxide is that current chemical kinetic models for this reactant system will have to be modified to account for high pressure effects. The modifications of these models will lead to a more accurate, comprehensive detailed chemical kinetic model that when coupled with computer design codes currently used by NASA will lead to the highest performance possible in the Mars vehicle rocket engine. The task objective is to use the unique single pulse shock tube at UIC to examine the oxidation of carbon monoxide with and without a hydrogen source over an extreme range of pressures and temperatures that encompass the design range of the Mars vehicle rocket engine so that recent models for carbon monoxide oxidation can be validated and made more comprehensive for application to the design of the high pressure rocket engine. To achieve the maximum performance from the Mars exploration vehicle rocket engine, which would be either pressure fed at 200-300 psi or pump fed at 2000-3000 psi, predictions or measurements of critical performance characteristics resulting from the use of carbon monoxide as the fuel are necessary. Among the critical characteristics are ignition of the combustible mixture, flame front heat release, and the effects of finite rate chemistry on specific impulse. These characteristics can be better predicted through an improved knowledge of the high pressure oxidation kinetics of carbon monoxide, as will be obtained in this task, and the better predictions necessary to achieve improved Mars rocket engine performance constitute the significance of the work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Carbon monoxide oxidation kinetics are an essential part of the modeling of the combustion of earth based hydrocarbon fuels such as gasoline, kerosene and jet propulsion fuels. Cleaner and more efficient combustion of these earth based fuels will come from better designs of combustion devices which have incorporated accurate models of hydrocarbon combustion into the design process. Therefore, an improved high pressure carbon monoxide oxidation model, as will come from this task work, will aid in the modeling and design of earth based combustion devices especially those operating at high pressures.
Task Progress:	Significant progress was made towards achieving the goals of the proposed study that involves: A: An Experimental and Modeling study of the High Pressure Oxidation of Carbon Monoxide. B: Obtaining a measurement of the “real” temperature in the shock tube. A: The Mars In-Situ Propellants Rocket (MIPR) is a small vehicle proposed to fly autonomously on Mars, using in-situ propellant production with the propulsion system being fueled by carbon monoxide and oxygen. There are detailed models available for carbon monoxide oxidation, but they have never been validated by experiments conducted at the operating pressures of the proposed Martian vehicle rocket engine, 200-300 psi (13,7-20.4 atm) for pressure fed engines, 2000-3000 psi (137-204 atm) for turbopump fed engines. Consequently the high pressure shock tube has been used in this work to study the combustion of CO/H2 mixtures over a wide range of pressures and temperatures relevant to the operating conditions of the proposed engine in order to develop a validated model that can determine the optimum design parameters for the proposed MIPR. Eight experimental data sets with more than 150 experiments spanning pressures from 25-550 bars over the temperature range 1000-1500 K with nominal reaction times of 1.2-1.8 ms were obtained. Five sets of experiments were performed for stoichiometric CO oxidation (F=1) spanning pressures from 25 to 550 bars and three sets of experiments at 25, 50 and 300 bars were performed using fuel lean (F=0.5) mixtures. The mixtures were doped with 160-225 ppm of H2. Standard GC/TCD-FID and GC/MS techniques were used to analyze the stable species (CO, CO2 and O2) from the shock samples. The experimental data was simulated using the recent Davis et al. model (An optimized model for CO/H2 combustion). Preliminary simulations indicate that the Davis et al. model is able to capture the trends for the lower pressure data sets (25 and 50 bars) but under predicts CO and O2 conversion and CO2 formation at high pressures. The modeling work is in progress and preliminary results suggest a modification in the fall-off parameters is warranted for the pressure dependent reaction H+O2+(Ar)=HO2(+Ar) in order to match the trends observed in the experiments. B: A fast response thermocouple (Type C) from NANMAC Corporation was found to be the best choice for obtaining the actual temperature within the shock tube. High frequency noise of amplitude of several orders of magnitude greater than the expected signal from the thermocouple was recorded by the data acquisition card during our preliminary tests and this resulted in the raw signals being masked noise. A detailed analysis of the source of the noise and methods to avoid the high frequency noise to be picked up by the data acquisition card was performed. The problems with the background noise have been resolved by the use of a differential amplifier as well as special electrical insulation (Kapton Tape) that was wrapped around all exposed parts of the thermocouple as well as the amplifier and currently experiments are in progress to obtain raw temperature measurements from the thermocouple.
Bibliography Type:	Description: (Last Updated: 02/21/2014)

Abstracts for Journals and Proceedings	Sivaramakrishnan R, Comandini A, Tranter RS, Brezinsky K. "High pressure experiments on the combustion of H2/CO mixtures." Conference PresentationSixth , Sixth International Conference on Chemical Kinetics, NIST, Gaitherburg, Maryland, July 24-29, 2005. International Conference on Chemical Kinetics, NIST, Gaitherburg, Maryland, July 2005. , Jul-2005
Dissertations and Theses	Comandini A. "High Pressure Study of Carbon Monoxide Oxidation for Application to Mars Exploration." M. S. Thesis, Department of Mechanical & Industrial Engineering, University of Illinois at Chicago, June 2005. , Jun-2005